Pistachio Species Classification Using Machine Learning: A Comparative Study of Seven Classifiers on Morphological Kernel Features

Sareena Bilal
sb743915@gmail.com

***Abstract*—** Pistachio (Pistacia vera L.) is a high-value nut crop whose commercial worth depends strongly on the variety, yet sorting by variety is still commonly performed by hand. This paper presents a reproducible machine-learning pipeline for distinguishing two Turkish pistachio varieties, Kirmizi and Siit, from sixteen morphological features extracted from kernel images. A dataset of 1,718 samples (998 Kirmizi, 720 Siit) was standardised and used to train and compare seven classical classifiers: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, k-Nearest Neighbours and Gaussian Naive Bayes. Models were assessed on an 80/20 hold-out split and, to obtain more reliable estimates, with leakage-free stratified 10-fold cross-validation. Gradient Boosting achieved the best overall performance, with a hold-out accuracy of 88.95% (AUC 0.952) and a cross-validated accuracy of 88.36% ± 2.31%, followed closely by the Support Vector Machine, Random Forest and Logistic Regression. The single Decision Tree was clearly the weakest model and overfitted the training data. Permutation importance analysis identified the minor-axis length of the kernel as by far the most discriminative feature. The results show that inexpensive shape descriptors combined with standard ensemble learners are sufficient for accurate, automated pistachio variety recognition.

***Keywords*—** *pistachio classification; machine learning; gradient boosting; morphological features; computer vision in agriculture; food quality inspection*

1. Introduction

Pistachio nuts are among the most economically important tree nuts in the world, with Iran, the United States and Türkiye accounting for most global production. Within Türkiye, the Kirmizi and Siit varieties are widely grown and traded, and they differ in market value, intended use and consumer preference. Correct identification of the variety is therefore relevant to pricing, packaging and quality control. In practice, separation is often done visually by trained workers, a process that is slow, subjective and difficult to scale to industrial throughput.

Computer vision combined with machine learning (ML) offers an objective alternative. Once a kernel has been photographed and segmented, simple geometric descriptors such as area, perimeter, axis lengths and shape factors can be computed automatically and fed to a classifier. This approach has been applied successfully to several agricultural products, including dry beans [10] and rice [11], and specifically to pistachios by Ozkan et al. [1] and Singh et al. [2].

This study builds on that line of work with a transparent, fully reproducible comparison of seven widely used classifiers on a public pistachio morphology dataset. The contributions are as follows:

- a systematic benchmark of seven classical classifiers under identical preprocessing and evaluation conditions;
- evaluation with both a hold-out split and stratified 10-fold cross-validation, reporting accuracy, precision, recall, specificity, F1-score, ROC-AUC and the Matthews correlation coefficient (MCC);
- an exploratory analysis of the feature space, including correlation structure and class-wise distributions, together with a permutation-based feature importance analysis; and

- openly available source code that allows every number and figure in this paper to be regenerated.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the dataset, preprocessing, classifiers and evaluation protocol. Section 4 presents the results, Section 5 discusses them along with the study's limitations, and Section 6 concludes.

2. Related Work

Ozkan, Koklu and Saracoglu [1] introduced the sixteen-feature pistachio dataset on which this study is based. Kernel images of the Kirmizi and Siit varieties were captured, segmented and described with morphological measurements, and an improved k-nearest-neighbour classifier combined with principal component analysis was reported to reach an accuracy of approximately 94%. Singh et al. [2] later revisited the problem from a deep-learning perspective, applying pre-trained convolutional neural networks (such as AlexNet, VGG16 and VGG19) directly to the pistachio images and reporting accuracies above 95%.

Similar feature-based pipelines have been used for other crops. Koklu and Ozkan [10] classified seven dry-bean varieties from shape features with multilayer perceptrons, support vector machines, k-NN and decision trees, while Cinar and Koklu [11] applied comparable methods to rice varieties. These studies consistently show that low-cost geometric descriptors carry enough information to separate visually similar varieties, and that kernel methods and ensembles tend to outperform single trees. The present work follows the same feature-based philosophy but focuses on a careful, like-for-like comparison of off-the-shelf scikit-learn [3] models with rigorous cross-validated evaluation.

3. Materials and Methods

Figure 1 summarises the overall workflow. Each stage is described in the subsections below.

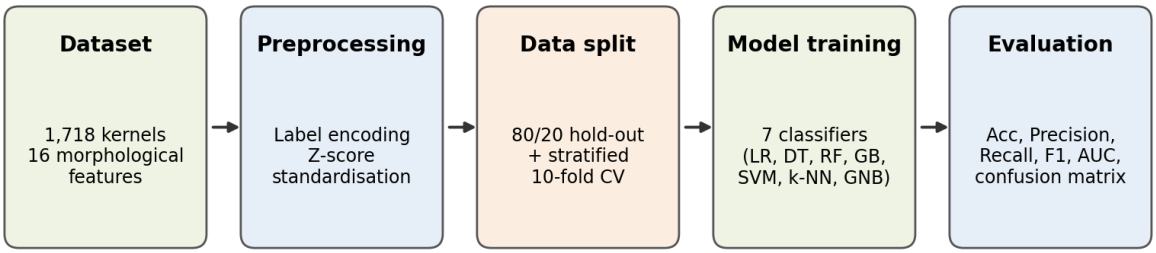


Fig. 1. Overview of the proposed pistachio classification pipeline.

3.1 Dataset

The dataset used in this study contains 1718 pistachio kernel samples belonging to two classes: Kirmizi (998 samples, 58.1%) and Siit (720 samples, 41.9%). It is the publicly redistributed, sixteen-feature version of the pistachio data originally described by Ozkan et al. [1]. Each sample is described by sixteen numerical morphological features computed from a segmented image of the kernel. The dataset contains no missing values. Table 1 defines each feature; the formulas were verified directly against the values in the data file.

Table 1. Morphological features used in this study (A = area, P = perimeter, L = major-axis length, l = minor-axis length).

Feature	Description / formula
AREA	Number of pixels inside the kernel boundary, A
PERIMETER	Length of the kernel boundary, P

Feature	Description / formula
MAJOR_AXIS	Length of the longest axis of the fitted ellipse, L
MINOR_AXIS	Length of the axis perpendicular to the major axis, l
ECCENTRICITY	Eccentricity of the fitted ellipse
EQDIASQ	Equivalent diameter, $\sqrt{(4A/\pi)}$
SOLIDITY	A / convex area
CONVEX_AREA	Pixel area of the smallest convex polygon containing the kernel
EXTENT	A / area of the bounding box
ASPECT_RATIO	L / l
ROUNDNESS	$4\pi A / P^2$
COMPACTNESS	EQDIASQ / L
SHAPEFACTOR_1	L / A
SHAPEFACTOR_2	l / A
SHAPEFACTOR_3	$A / (\pi(L/2)^2)$
SHAPEFACTOR_4	$A / (\pi(L/2)(l/2))$

Table 2 reports the mean and standard deviation of every feature for each class. Siit kernels are, on average, larger (higher area and minor-axis length) and rounder (lower eccentricity and aspect ratio, higher compactness) than Kirmizi kernels, which already suggests that size- and shape-related features will be informative.

Table 2. Class-wise descriptive statistics (mean ± standard deviation).

Feature	Kirmizi (n = 998)	Siit (n = 720)
AREA	73313 ± 10885	88963 ± 9739
PERIMETER	1373 ± 333	1489 ± 414
MAJOR_AXIS	442.25 ± 34.32	451.70 ± 27.25
MINOR_AXIS	221.00 ± 22.30	262.02 ± 23.29
ECCENTRICITY	0.8629 ± 0.0343	0.8091 ± 0.0493
EQDIASQ	304.62 ± 23.54	336.05 ± 18.52
SOLIDITY	0.9350 ± 0.0494	0.9471 ± 0.0500
CONVEX_AREA	78390 ± 10921	94037 ± 10044
EXTENT	0.7115 ± 0.0543	0.7224 ± 0.0493
ASPECT_RATIO	2.0166 ± 0.2098	1.7368 ± 0.1801
ROUNDNESS	0.5520 ± 0.1934	0.5969 ± 0.2319
COMPACTNESS	0.6896 ± 0.0332	0.7451 ± 0.0367
SHAPEFACTOR_1	0.00613 ± 0.00075	0.00512 ± 0.00045
SHAPEFACTOR_2	0.00305 ± 0.00035	0.00297 ± 0.00030
SHAPEFACTOR_3	0.4766 ± 0.0453	0.5565 ± 0.0551
SHAPEFACTOR_4	0.9528 ± 0.0484	0.9585 ± 0.0559

3.2 Preprocessing

The class label was converted to a binary integer with label encoding (Kirmizi = 0, Siit = 1); Siit is therefore treated as the positive class in all metrics. Because the features span very different numerical ranges (for example, AREA is of the order of 10^5 whereas SHAPEFACTOR_2 is of the order of 10^{-3}), all features were standardised to zero mean and unit variance using z-score scaling:

$$z = (x - \mu) / \sigma \qquad (1)$$

where μ and σ are the mean and standard deviation of the feature. Standardisation is essential for distance- and margin-based models such as k-NN, SVM and Logistic Regression, and is harmless for tree-based models. In the hold-out experiment the scaler follows the original implementation and is fitted on the full feature matrix before splitting. In the cross-validation experiment the scaler is embedded in a scikit-learn pipeline so that it is refitted on the training folds only, eliminating any information leakage from the validation folds.

3.3 Data Partitioning

Two evaluation protocols were used. First, the data were divided into a training set (80%, 1,374 samples) and a test set (20%, 344 samples: 201 Kirmizi and 143 Siit) with a fixed random seed of 42, reproducing the split of the original project. Second, because a single split can give an optimistic or pessimistic picture by chance, every model was also evaluated with stratified 10-fold cross-validation (shuffled, seed 42), which preserves the class ratio in each fold and yields both a mean and a standard deviation for every metric [15].

3.4 Classifiers

Seven supervised classifiers, representing linear, tree-based, ensemble, kernel, instance-based and probabilistic families, were compared. All were implemented with scikit-learn [3] using default hyperparameters (Table 3) so that the comparison reflects each algorithm's out-of-the-box behaviour.

- **Logistic Regression (LR)** [9] models the log-odds of the positive class as a linear function of the features and serves as a strong linear baseline.
- **Decision Tree (DT)** [8] recursively partitions the feature space using axis-aligned splits that maximise the reduction in Gini impurity.
- **Random Forest (RF)** [4] averages many decorrelated trees trained on bootstrap samples with random feature subsets, reducing the variance of a single tree.
- **Gradient Boosting (GB)** [5] builds an additive ensemble of shallow trees in which each new tree fits the residual errors of the current model.
- **Support Vector Machine (SVM)** [6] finds the maximum-margin separating hyperplane in a feature space induced by a radial basis function (RBF) kernel.
- **k-Nearest Neighbours (k-NN)** [7] assigns the majority class among the k closest training samples in Euclidean distance.
- **Gaussian Naive Bayes (GNB)** applies Bayes' theorem under the assumption that features are conditionally independent and normally distributed within each class.

Table 3. Classifier configurations (scikit-learn defaults unless stated).

Classifier	Key hyperparameters
Logistic Regression	L2 penalty, C = 1.0, lbfgs solver
Decision Tree	Gini criterion, unlimited depth, min_samples_split = 2
Random Forest	100 trees, Gini, max_features = √d, bootstrap
Gradient Boosting	100 estimators, learning rate 0.1, max_depth = 3, log-loss
Support Vector Machine	RBF kernel, C = 1.0, γ = "scale"
k-Nearest Neighbours	k = 5, uniform weights, Euclidean distance
Gaussian Naive Bayes	var_smoothing = 1e-9

3.5 Evaluation Metrics

Let TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives with Siit as the positive class. The following metrics were computed:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (2)$$

$$Precision = TP / (TP + FP) \quad (3)$$

$$Recall\ (Sensitivity) = TP / (TP + FN) \quad (4)$$

$$Specificity = TN / (TN + FP) \quad (5)$$

$$F1 = 2 \cdot Precision \cdot Recall / (Precision + Recall) \quad (6)$$

$$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{((TP+FP)(TP+FN)(TN+FP)(TN+FN))} \quad (7)$$

In addition, the area under the receiver operating characteristic curve (ROC-AUC) was computed from predicted class probabilities. The MCC is included because it remains informative under class imbalance and uses all four cells of the confusion matrix [12]. Finally, permutation feature importance [4] was computed for the best model by measuring the mean drop in test accuracy over 30 random shuffles of each feature.

4. Results

4.1 Exploratory Data Analysis

Figure 2 shows the class distribution. The dataset is moderately imbalanced (58.1% versus 41.9%), so a trivial majority-class predictor would achieve 58.1% accuracy; this is the baseline against which all models should be judged.

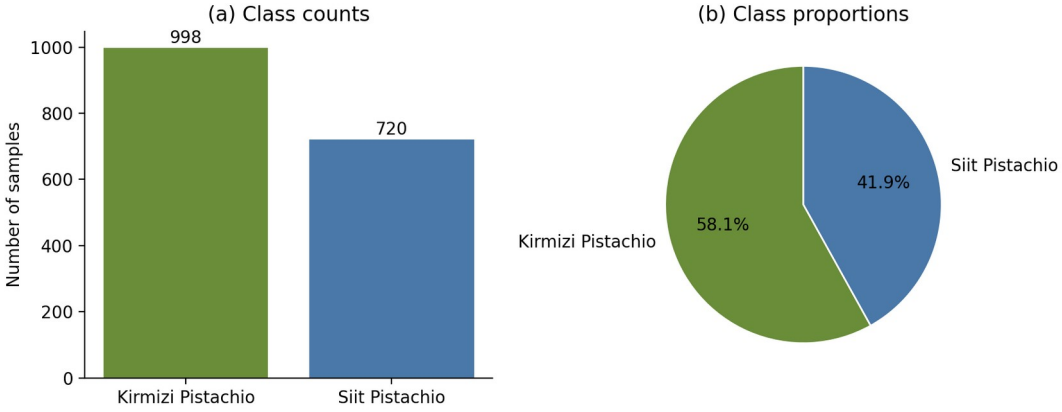


Fig. 2. Class distribution of the pistachio dataset: (a) absolute counts and (b) proportions.

The Pearson correlation matrix in Figure 3 reveals strong redundancy among the features. AREA and EQDIASQ are almost perfectly correlated (r = 0.997) because the equivalent diameter is a monotonic function of area, and COMPACTNESS and SHAPEFACTOR_3 are nearly identical (r = 0.999) because SHAPEFACTOR_3 equals the square of compactness. AREA, CONVEX_AREA and MINOR_AXIS also form a highly correlated size cluster. Such multicollinearity does not harm predictive accuracy for regularised or tree-based models, but it does mean that the effective dimensionality of the data is well below sixteen.

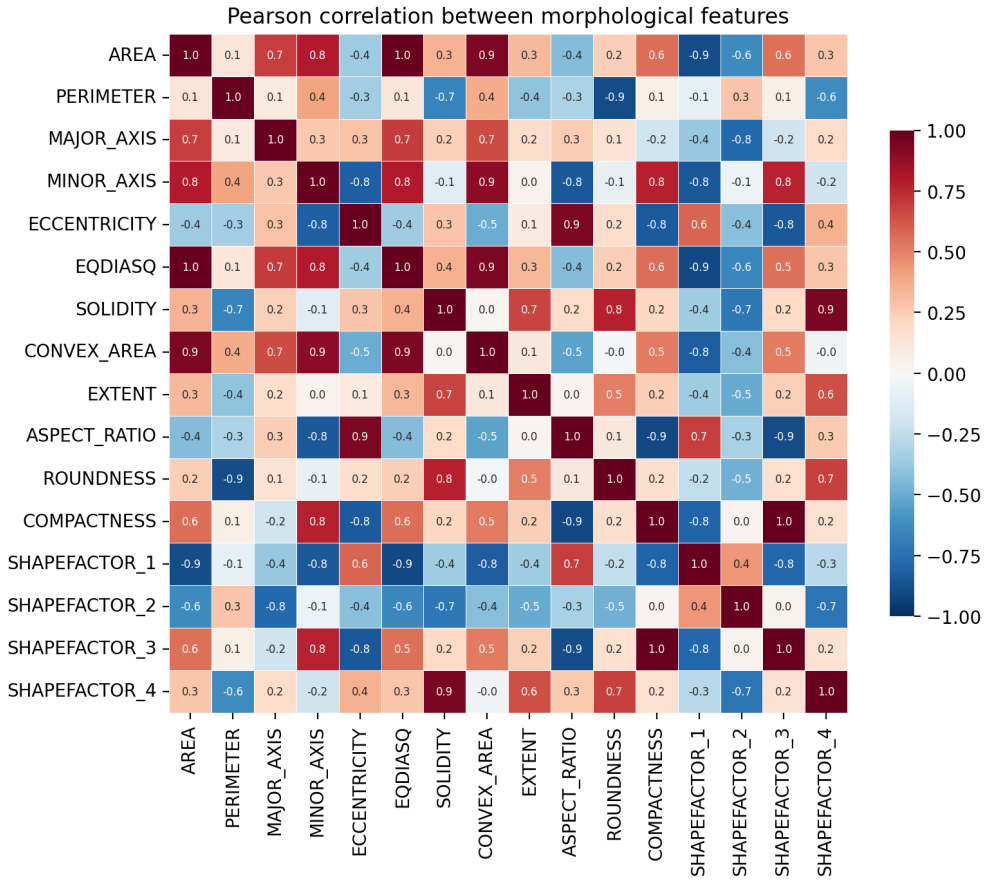


Fig. 3. Pearson correlation matrix of the sixteen morphological features.

Figure 4 compares the class-wise distributions of six representative features. AREA, ECCENTRICITY and ASPECT_RATIO show a clear shift between the two varieties, while SOLIDITY and SHAPEFACTOR_4 overlap heavily and contain many low-valued outliers, most likely caused by kernels with open shells or irregular contours.

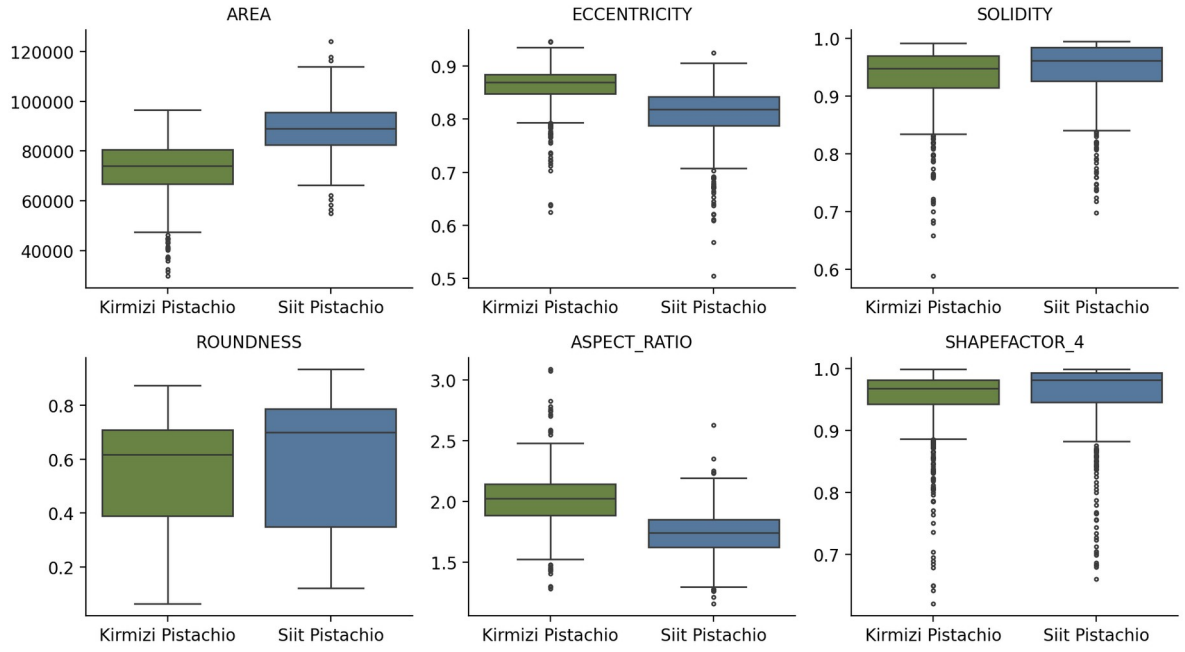


Fig. 4. Class-wise box plots of six representative morphological features.

4.2 Hold-out Test Results

Table 4 reports the performance of each classifier on the 344-sample hold-out test set, and Figure 5 shows the corresponding confusion matrices. Every model comfortably exceeds the 58.1% majority baseline. Gradient Boosting obtained the highest accuracy (88.95%), precision (0.8889), specificity (0.9254), F1-score (0.8633), AUC (0.9516) and MCC (0.7717). Logistic Regression and the SVM followed closely, with accuracies of 88.37% and 88.08%. Gaussian Naive Bayes achieved the highest recall (0.8951) but at the cost of the second-lowest precision, meaning it tends to over-predict the Siit class. The Decision Tree was the weakest model on every metric.

Table 4. Hold-out test performance (n = 344). The best value in each column is marked with *; Spec. = specificity.

Classifier	Accuracy	Precision	Recall	Spec.	F1	AUC	MCC
Logistic Regression	0.8837	0.8552	0.8671	0.8955	0.8611	0.9419	0.7612
Decision Tree	0.8052	0.7714	0.7552	0.8408	0.7633	0.7980	0.5979
Random Forest	0.8750	0.8521	0.8462	0.8955	0.8491	0.9381	0.7424
Gradient Boosting	**0.8895 ***	**0.8889 ***	**0.8392**	**0.9254 ***	**0.8633 ***	**0.9516 ***	**0.7717 ***
Support Vector Machine	0.8808	0.8493	0.8671	0.8905	0.8581	0.9487	0.7555
K-Nearest Neighbors	0.8634	0.8333	0.8392	0.8806	0.8362	0.9173	0.7190
Gaussian Naive Bayes	0.8721	0.8153	0.8951 *	0.8557	0.8533	0.9417	0.7429

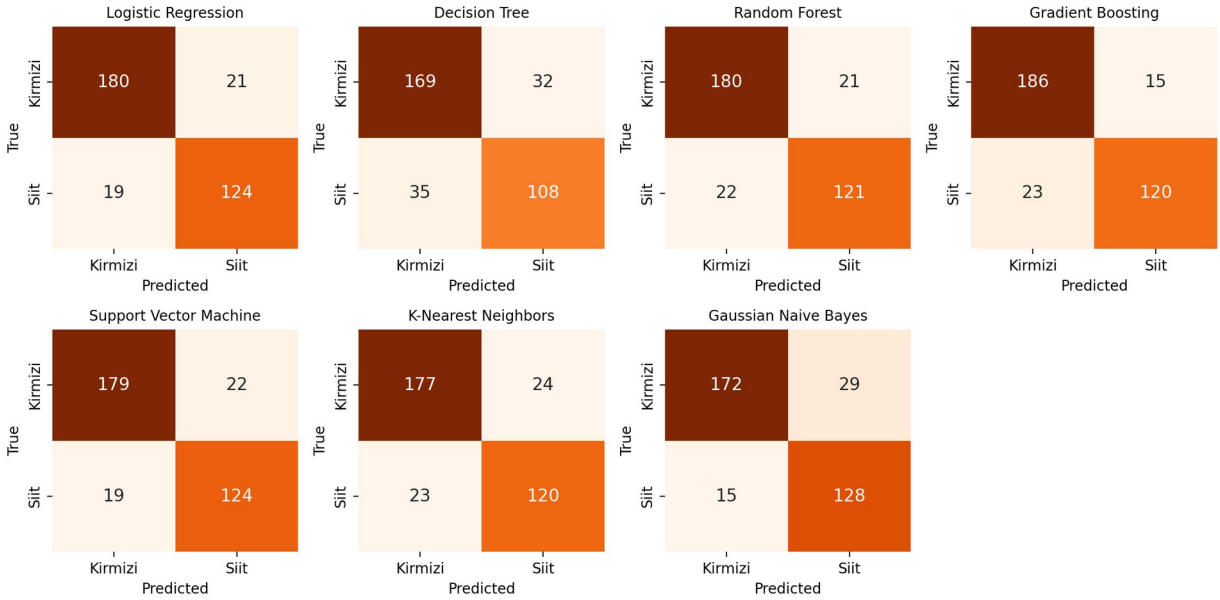


Fig. 5. Confusion matrices of the seven classifiers on the hold-out test set (rows: true class; columns: predicted class).

Figure 6 visualises the four threshold-dependent metrics side by side, and Figure 7 shows the ROC curves. All ensemble, kernel and linear models achieve AUC values between 0.92 and 0.95, indicating good ranking ability across thresholds, whereas the Decision Tree, which outputs nearly hard probabilities, yields an AUC of only 0.798.

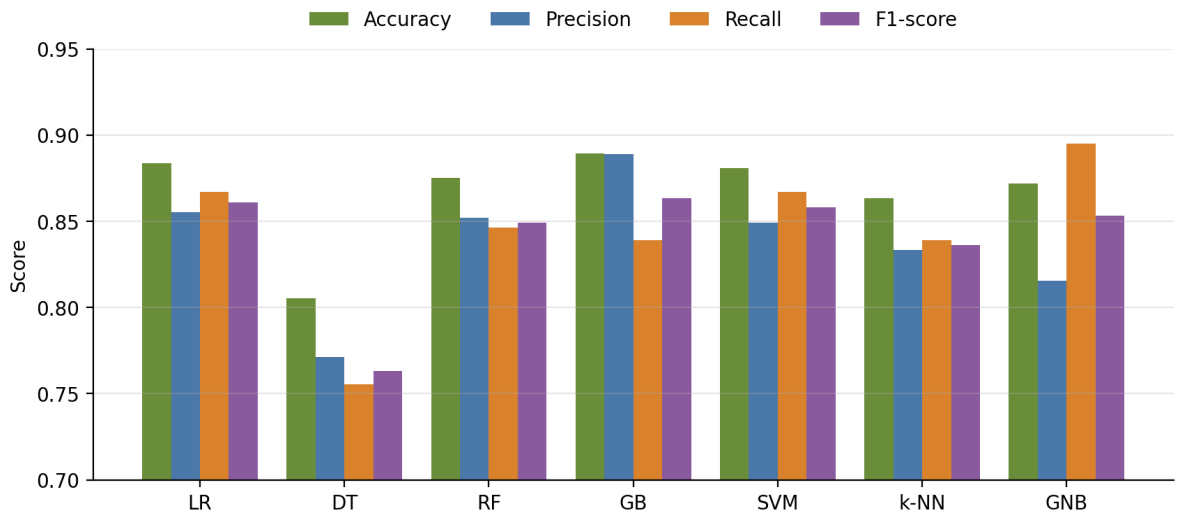


Fig. 6. Comparison of accuracy, precision, recall and F1-score across classifiers on the hold-out test set.

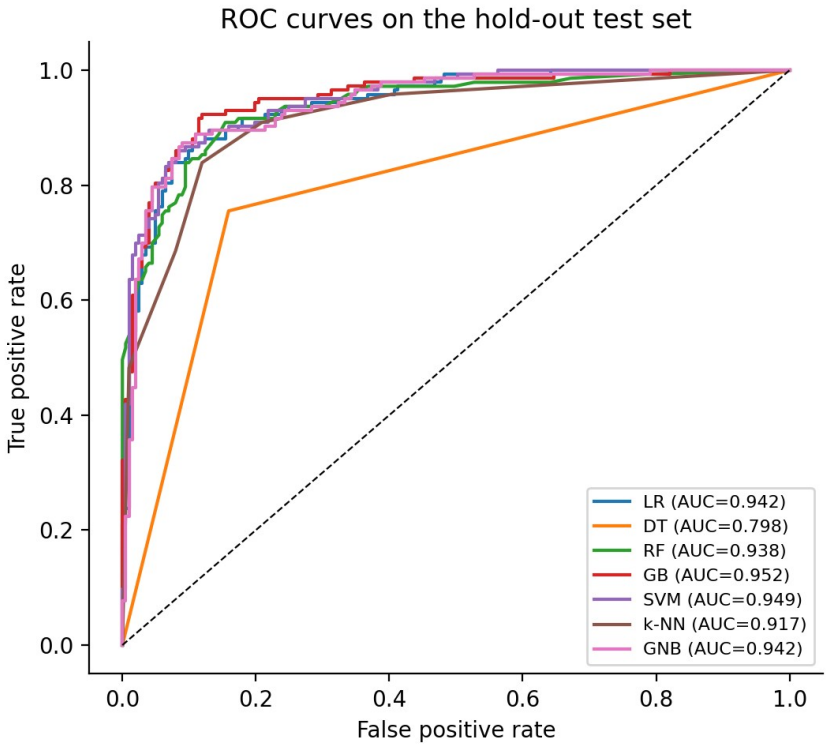


Fig. 7. ROC curves and AUC values on the hold-out test set.

Figure 8 contrasts training and test accuracy. The Decision Tree and Random Forest reach 100% training accuracy, but only the Random Forest generalises well, because averaging many trees cancels much of their individual variance. The unpruned single tree shows a gap of almost 20 percentage points, a classic sign of overfitting. Gradient Boosting shows a moderate gap (94.61% versus 88.95%), while LR, SVM and GNB have nearly equal training and test accuracies, indicating low variance.

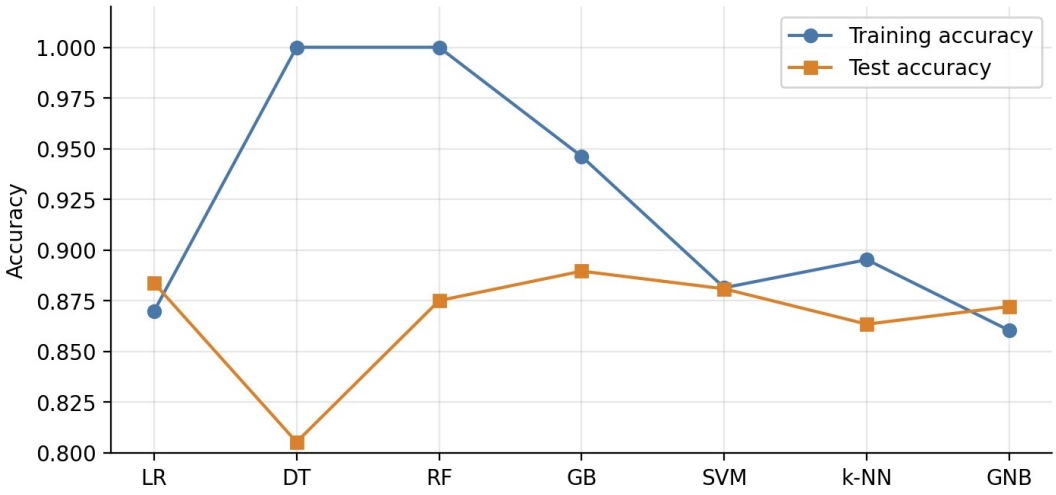


Fig. 8. Training versus test accuracy for each classifier.

4.3 Cross-Validation Results

Table 5 and Figure 9 summarise the stratified 10-fold cross-validation results, in which scaling was performed inside each fold. The ranking largely confirms the hold-out experiment: Gradient Boosting again achieved the best mean accuracy (88.36%), F1-score and AUC, with the smallest spread of F1 across folds, followed by the SVM (87.78%) and Random Forest (87.49%). The differences among the top five models are within roughly one to two standard deviations, so they should be regarded as broadly comparable; only the Decision Tree is clearly and consistently worse.

Table 5. Stratified 10-fold cross-validation results (mean ± standard deviation). Best model shown in bold.

Classifier	Accuracy (%)	Precision	Recall	F1	AUC
Logistic Regression	86.91 ± 2.97	0.8533 ± 0.0426	0.8319 ± 0.0423	0.8418 ± 0.0358	0.9395 ± 0.0211
Decision Tree	81.26 ± 2.80	0.7751 ± 0.0311	0.7792 ± 0.0503	0.7765 ± 0.0360	0.8079 ± 0.0303
Random Forest	87.49 ± 2.25	0.8581 ± 0.0314	0.8417 ± 0.0408	0.8491 ± 0.0282	0.9388 ± 0.0206
Gradient Boosting	**88.36 ± 2.31**	**0.8676 ± 0.0386**	**0.8542 ± 0.0243**	**0.8604 ± 0.0259**	**0.9446 ± 0.0212**
Support Vector Machine	87.78 ± 2.95	0.8646 ± 0.0425	0.8417 ± 0.0399	0.8524 ± 0.0349	0.9363 ± 0.0210
K-Nearest Neighbors	86.61 ± 2.58	0.8508 ± 0.0427	0.8278 ± 0.0312	0.8385 ± 0.0294	0.9197 ± 0.0221
Gaussian Naive Bayes	86.50 ± 2.86	0.8268 ± 0.0361	0.8583 ± 0.0434	0.8418 ± 0.0344	0.9350 ± 0.0227

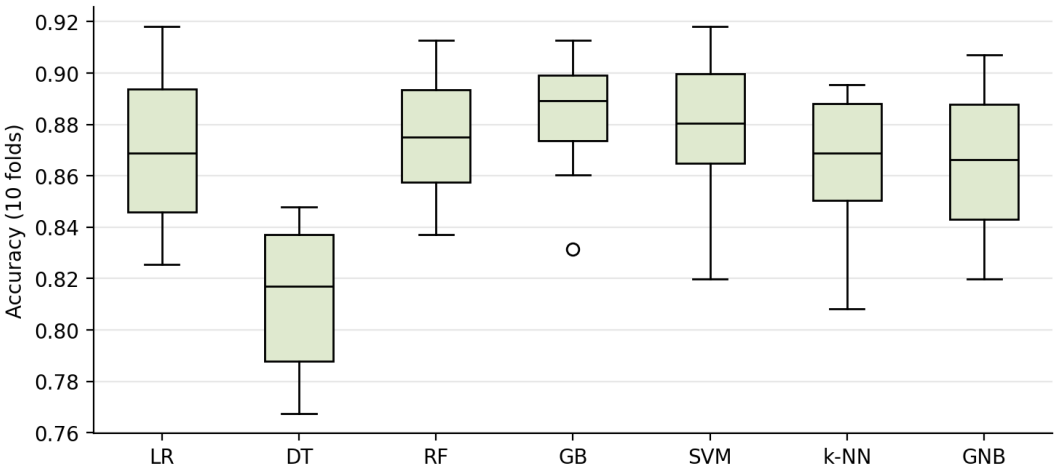


Fig. 9. Distribution of fold accuracies in stratified 10-fold cross-validation.

4.4 Feature Importance

Figure 10 shows the permutation importance of each feature for the Gradient Boosting model. MINOR_AXIS dominates: shuffling it reduces test accuracy by about 16.5 percentage points, several times more than any other feature. It is followed by SHAPEFACTOR_1, SHAPEFACTOR_4, SOLIDITY and ROUNDNESS. Several features have importance close to zero or slightly negative. This does not mean they are uninformative in isolation (AREA, for instance, differs markedly between classes, Table 2); rather, because they are highly correlated with features the model already uses, permuting them alone removes little information. This confirms the redundancy observed in Figure 3 and suggests that a much smaller feature subset could achieve similar accuracy.

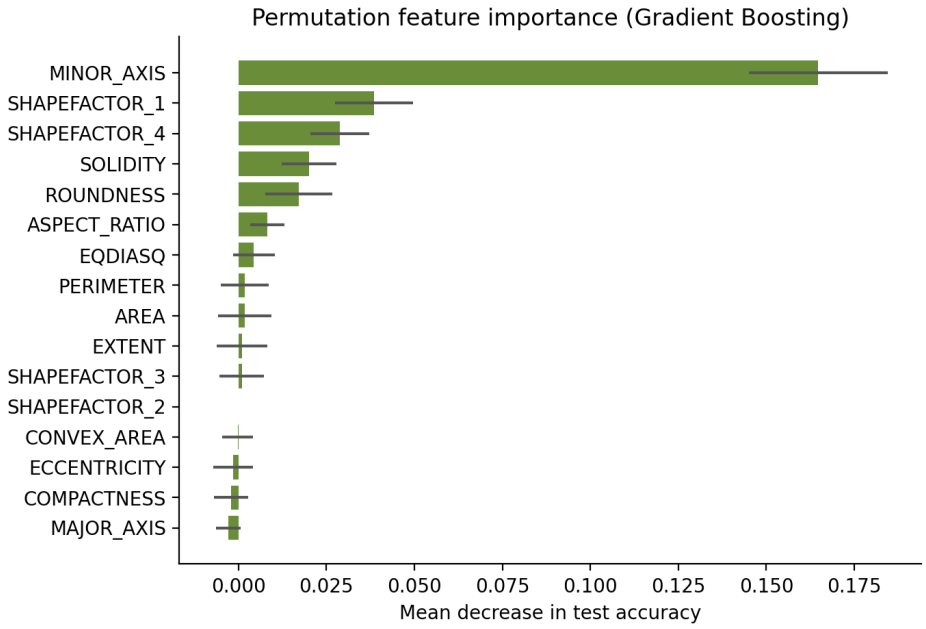


Fig. 10. Permutation feature importance for the Gradient Boosting classifier (mean ± standard deviation over 30 repeats).

4.5 Reproducibility Check

Table 6 compares the test accuracies recorded in the original project notebook with those obtained when the pipeline was re-executed for this paper. The deterministic models (LR, SVM, k-NN, GNB) reproduce exactly. Small differences for DT, RF and GB arise because the original code did not fix the random seed of these models; in this study, random_state = 42 was set for all stochastic estimators so that the results are fully repeatable. Gradient Boosting is the best model in both runs.

Table 6. Hold-out accuracy recorded in the original notebook versus the re-executed, seeded pipeline.

Classifier	Original notebook	This study (seeded)	Difference
Logistic Regression	0.8837	0.8837	0.0000
Decision Tree	0.8140	0.8052	−0.0088
Random Forest	0.8634	0.8750	+0.0116
Gradient Boosting	0.8924	0.8895	−0.0029
Support Vector Machine	0.8808	0.8808	0.0000
K-Nearest Neighbors	0.8634	0.8634	0.0000
Gaussian Naive Bayes	0.8721	0.8721	0.0000

5. Discussion

The experiments show that two Turkish pistachio varieties can be distinguished with roughly 88–89% accuracy using only sixteen inexpensive shape measurements and standard, untuned classifiers. Ensemble methods, in particular Gradient Boosting, gave the best and most stable results, which is consistent with findings on other crop-classification tasks [10], [11]. Gradient Boosting combines many shallow trees, which allows it to capture the non-linear interactions between size and shape features while avoiding the high variance of a single deep tree.

The strong performance of Logistic Regression is also notable. That a linear model trails the best ensemble by only around one percentage point suggests that the two classes are close to linearly separable in the standardised feature space, and that most of the remaining errors stem from genuine overlap between the varieties rather than from model capacity. The confusion matrices support this view: most models make similar numbers of errors on each class, and many misclassified samples are shared across models.

The accuracies reported here are lower than the approximately 94% reported by Ozkan et al. [1] with an improved k-NN and PCA, and the over 95% reported with deep convolutional networks on the raw images [2]. Several factors may explain this gap: the present study uses default hyperparameters, a smaller redistributed version of the dataset, and features only rather than images. The results should therefore be interpreted as a strong, transparent baseline rather than a new state of the art.

Limitations. First, no hyperparameter optimisation was performed; grid or Bayesian search over, for example, tree depth, number of estimators, SVM C and γ, or k in k-NN would probably improve all models. Second, in the hold-out experiment the scaler was fitted before splitting, as in the original code; the effect on accuracy is negligible here, and the cross-validation experiment avoids it entirely, but a deployment pipeline should always fit preprocessing on training data alone. Third, the dataset covers only two varieties from a single region and imaging set-up, so generalisation to other cultivars, cameras or lighting conditions remains untested. Finally, the strong multicollinearity among features was not addressed; feature selection or PCA could yield a simpler and more interpretable model.

6. Conclusion and Future Work

This paper presented a reproducible comparison of seven machine-learning classifiers for distinguishing Kirmizi and Siit pistachios from morphological kernel features. Gradient Boosting achieved the best performance, with 88.95% hold-out accuracy, an AUC of 0.952 and 88.36% mean cross-validated accuracy, while the SVM, Random Forest and Logistic Regression were close behind. The minor-axis length emerged as the most discriminative single feature, and the feature set as a whole was found to be highly redundant.

Future work will focus on hyperparameter tuning, feature selection and dimensionality reduction, the evaluation of modern boosting libraries and neural networks, and the combination of hand-crafted shape features with texture and colour descriptors or deep image embeddings. Validating the approach on larger, multi-variety datasets and integrating the trained model into a real-time sorting system are also promising directions.

Code and Data Availability

The complete source code, Jupyter notebook and dataset used in this study are available at: https://github.com/Sareena-2001/PISTACHIO-CLASSIFICATION-USING-MACHINE-LEARNING

References

[1] I. A. Ozkan, M. Koklu, and R. Saracoglu, “Classification of pistachio species using improved k-NN classifier,” Progress in Nutrition, vol. 23, no. 2, e2021044, 2021.

[2] D. Singh, Y. S. Taspinar, R. Kursun, I. Cinar, M. Koklu, I. A. Ozkan, and H.-N. Lee, “Classification and analysis of pistachio species with pre-trained deep learning models,” Electronics, vol. 11, no. 7, p. 981, 2022.

[3] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[4] L. Breiman, “Random forests,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[5] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” The Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001.

[6] C. Cortes and V. Vapnik, “Support-vector networks,” Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[7] T. Cover and P. Hart, “Nearest neighbor pattern classification,” IEEE Transactions on Information Theory, vol. 13, no. 1, pp. 21–27, 1967.

[8] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, Classification and Regression Trees. Belmont, CA, USA: Wadsworth, 1984.

[9] D. R. Cox, “The regression analysis of binary sequences,” Journal of the Royal Statistical Society: Series B, vol. 20, no. 2, pp. 215–242, 1958.

[10] M. Koklu and I. A. Ozkan, “Multiclass classification of dry beans using computer vision and machine learning techniques,” Computers and Electronics in Agriculture, vol. 174, p. 105507, 2020.

[11] I. Cinar and M. Koklu, “Classification of rice varieties using artificial intelligence methods,” International Journal of Intelligent Systems and Applications in Engineering, vol. 7, no. 3, pp. 188–194, 2019.

[12] D. Chicco and G. Jurman, “The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation,” BMC Genomics, vol. 21, p. 6, 2020.

[13] J. D. Hunter, “Matplotlib: A 2D graphics environment,” Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.

[14] M. L. Waskom, “seaborn: Statistical data visualization,” Journal of Open Source Software, vol. 6, no. 60, p. 3021, 2021.

[15] R. Kohavi, “A study of cross-validation and bootstrap for accuracy estimation and model selection,” in Proc. 14th International Joint Conference on Artificial Intelligence (IJCAI), 1995, pp. 1137–1145.